CISPA
HELMHOLTZ CENTER FOR INFORMATION SECURITY

Cryptography

Lecture 07:Block Ciphers

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Pseudorandom Functions

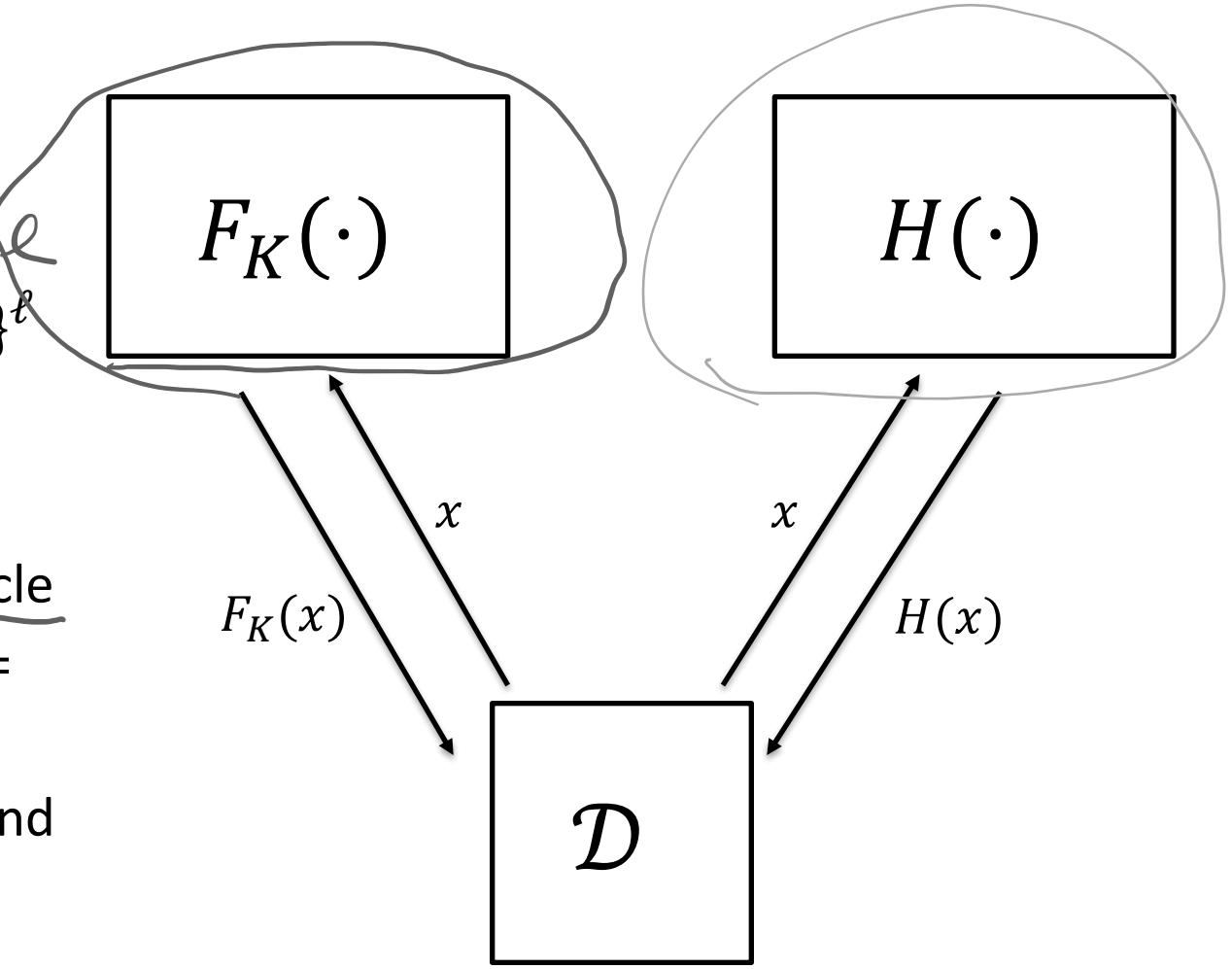
$Enc_K(m;r) = (r, PRF_K(r) \oplus m)$

Definition

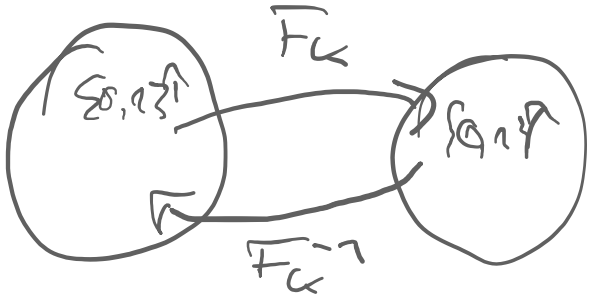
- Efficiently computable keyed function (ensemble) $F_K: \{0,1\}^\lambda \to \{0,1\}^\ell$, where $K \in \{0,1\}^\lambda$

  Signature

  description size $2^\lambda \cdot \ell$
- Uniformly random function $H: \{0,1\}^\lambda \to \{0,1\}^\ell$
- i.e. for every $x \in \{0,1\}^\lambda$ it holds that $H(x)$ is independently uniformly random on $\{0,1\}^\ell$
- F is a pseudorandom function, if for every oracle PPT distinguisher $\mathcal{D}$ it holds $|Pr[\mathcal{D}^{F_K(\cdot)}(1^\lambda) = 1] - Pr[\mathcal{D}^{H(\cdot)}(1^\lambda) = 1]| < negl(\lambda)$ where probability is over choice of $K \leftarrow_\$ \{0,1\}^\lambda$, H and random coins of $\mathcal{D}$



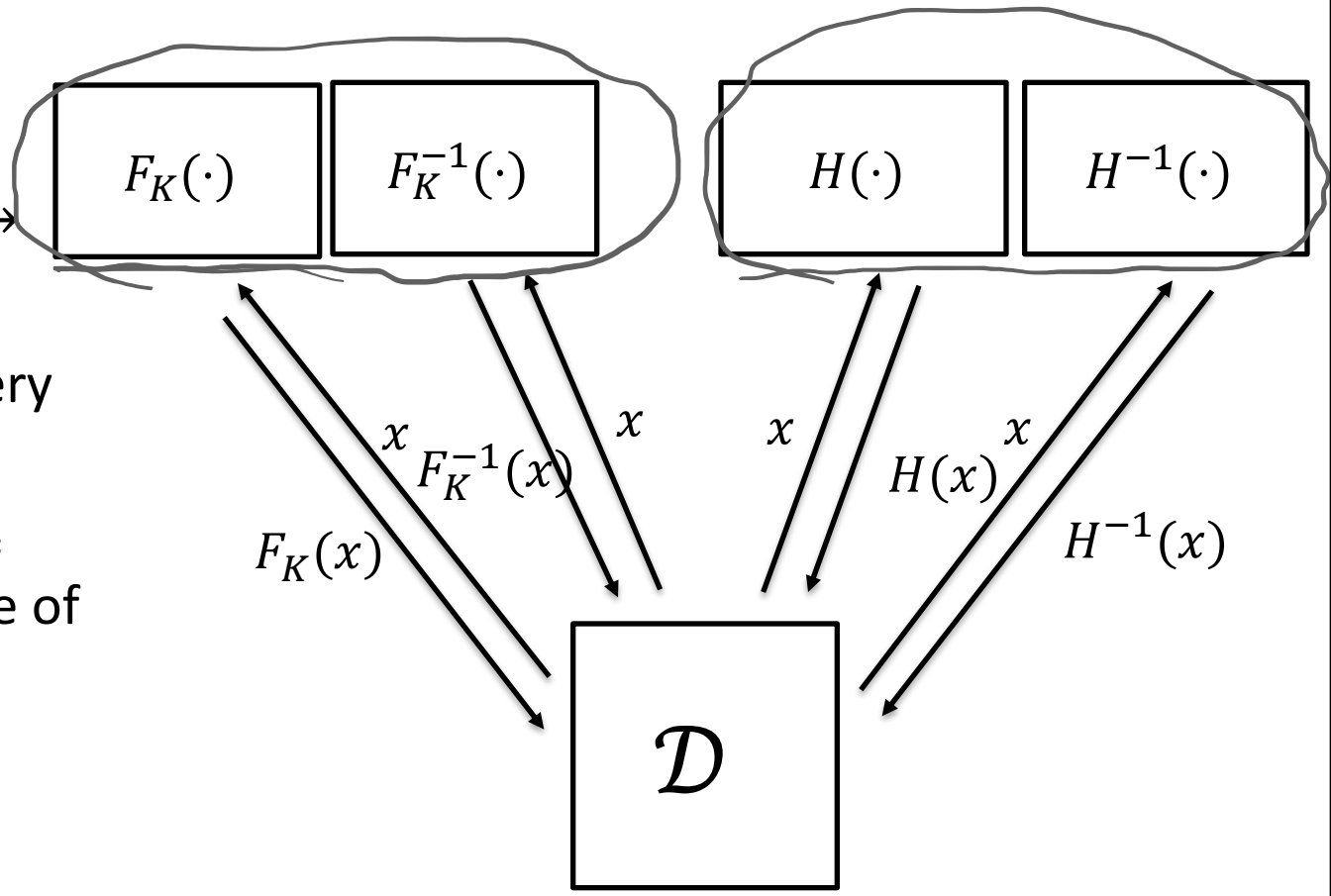
Pseudorandom Permutations (PRPs)



$F_K(x)$

Definition

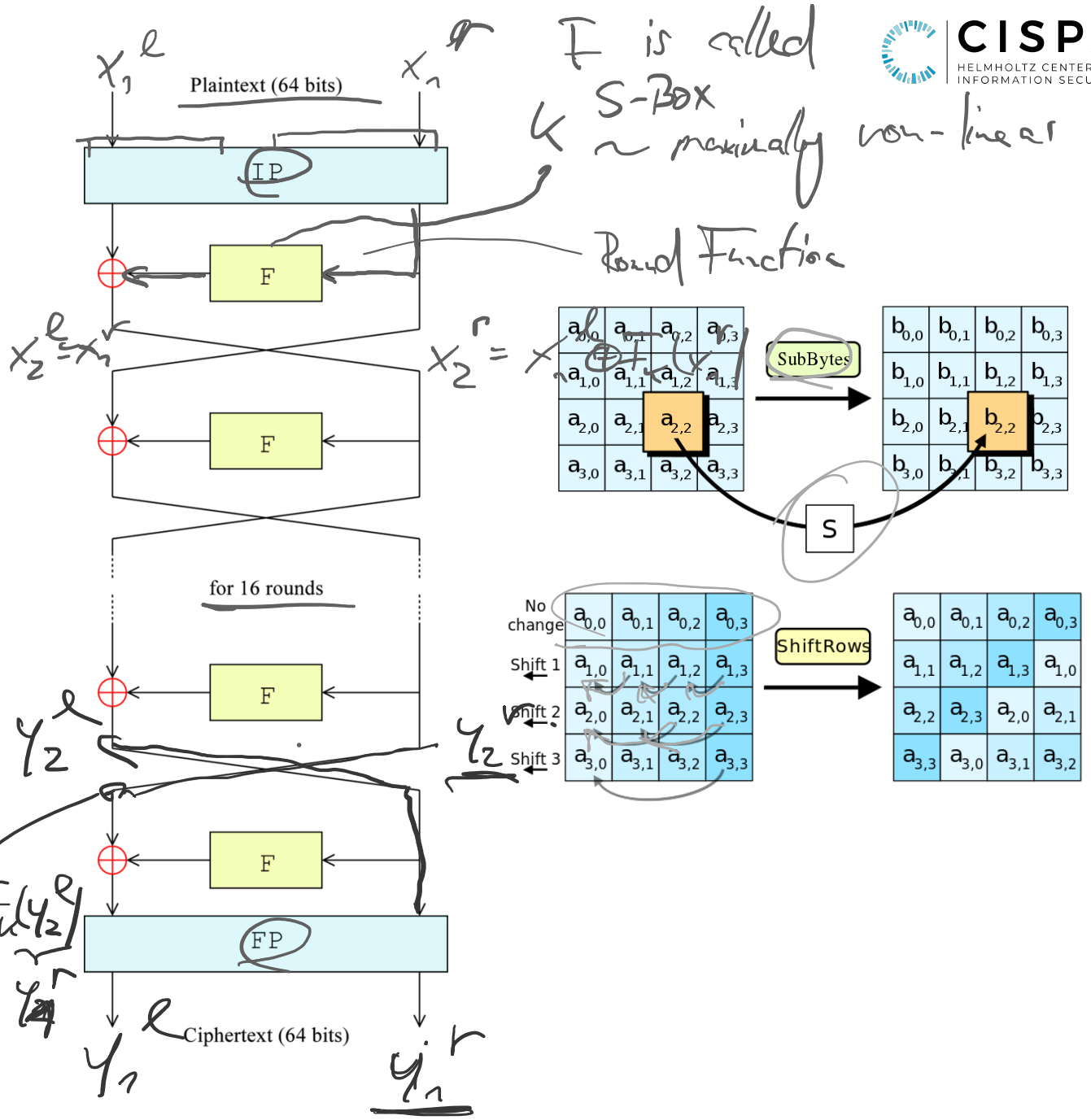
- Efficiently computable keyed function (ensemble) $F_K: \{0,1\}^\lambda \to \{0,1\}^\lambda$ and $F_K^{-1}: \{0,1\}^\lambda \to \{0,1\}^\lambda$, where $K \in \{0,1\}^\lambda$
- Uniformly random permutation $H: \{0,1\}^\lambda \to \{0,1\}^\lambda$
- F is a pseudorandom permutation, if for every oracle PPT distinguisher $\mathcal{D}$ it holds $|Pr[\mathcal{D}^{F_K(\cdot),F_K^{-1}(\cdot)}(1^\lambda) = 1] - Pr[\mathcal{D}^{H(\cdot),H^{-1}(\cdot)}(1^\lambda) = 1]| < negl(\lambda)$ where probability is over choice of $K \leftarrow_\$ \{0,1\}^\lambda$, H and random coins of $\mathcal{D}$



Block ciphers

- Practitioners refer to pseudorandom permutations as **block ciphers**
- Practical constructions of block ciphers follow **best practices** and are **guided by cryptanalysis**
- I.e. practical constructions of block ciphers typically do not come with a security reduction
- Popular design patterns:
  - Feistel Schemes (Data Encryption Standard/DES)
  - Substitution-Permutation-Networks (Advanced Encryption Standard/AES)

Rijndael

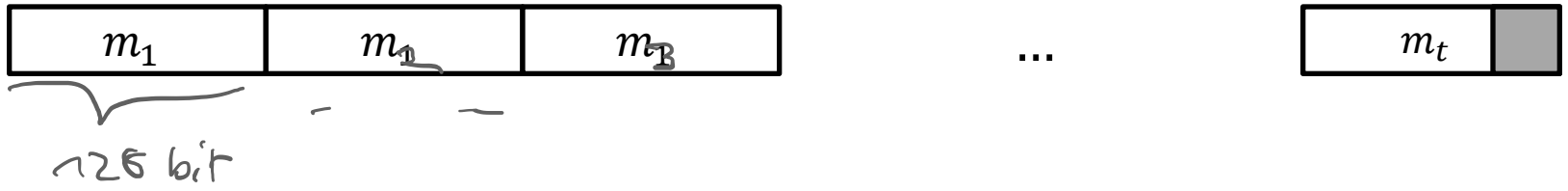


$\Leftrightarrow y_2^r = y_1^l \oplus F_k(y_1^r)$

Using Block-Ciphers

- Block-ciphers/PRPs are a *primitive*
- How do we construct encryption from PRPs?
- Constructions of encryption from block-ciphers are called **modes of operation**
- Assume messages can be chopped into blocks of size λ, otherwise use padding

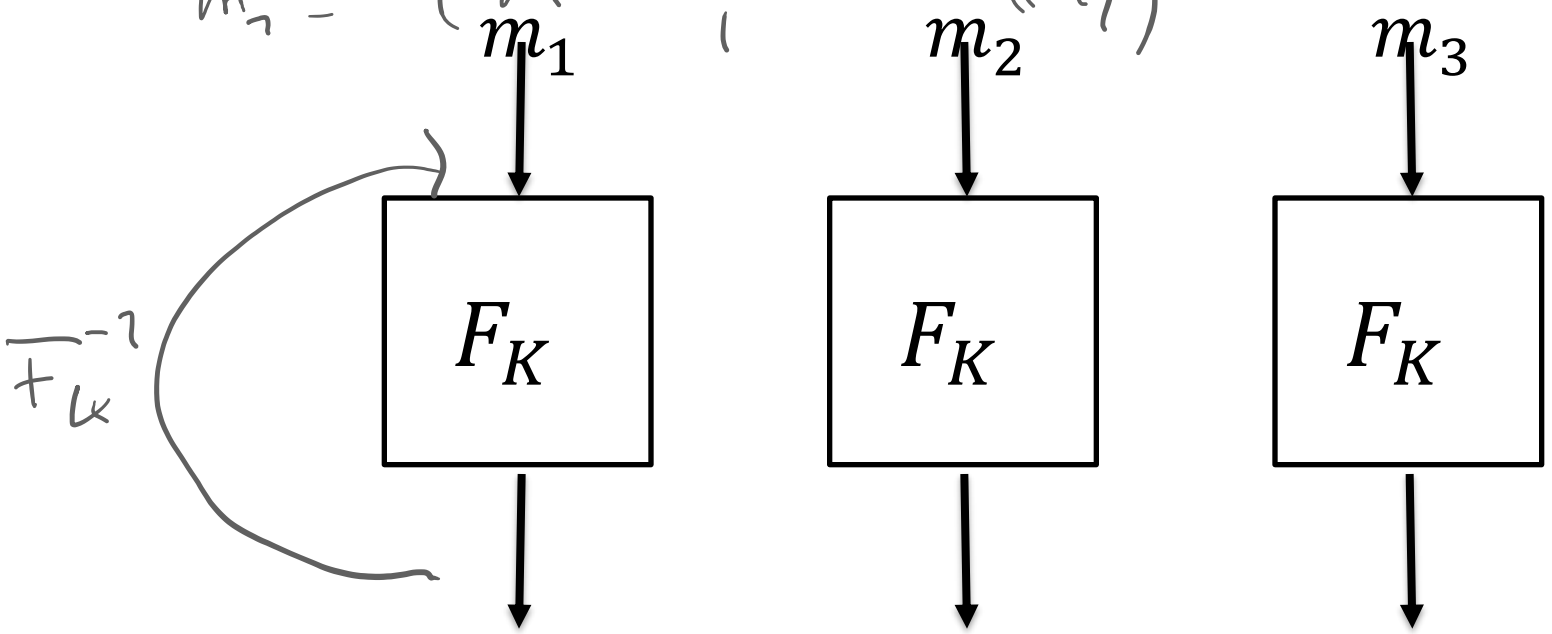
$m =$



Electronic Codebook Mode (ECM)

$m_0 = (m, m)$

$m_1 = (m, m' (\neq m))$



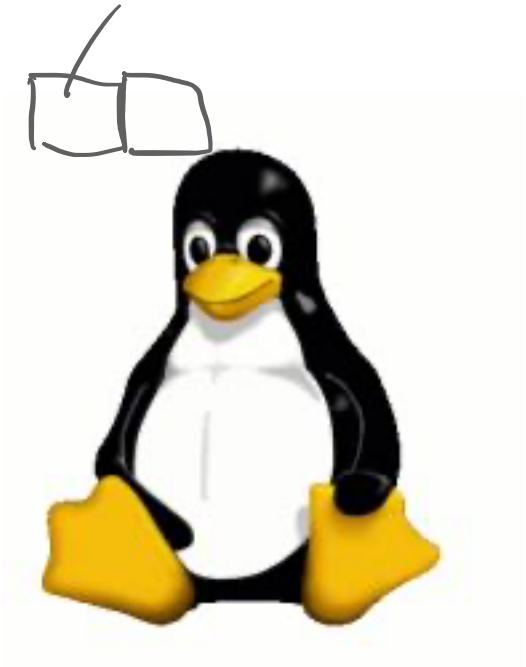
$m_0 \rightarrow c$

$m_1 \rightarrow c'$

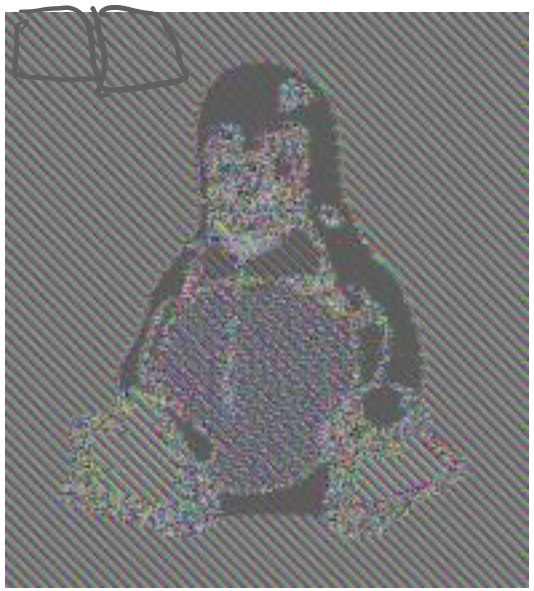
- Not IND-CPA secure!
- Not even IND secure!
- Should not be used!

Electronic Codebook Mode (ECM)

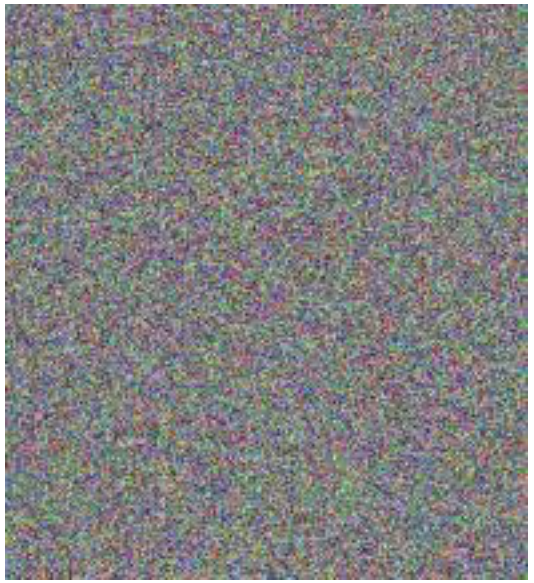
16x16



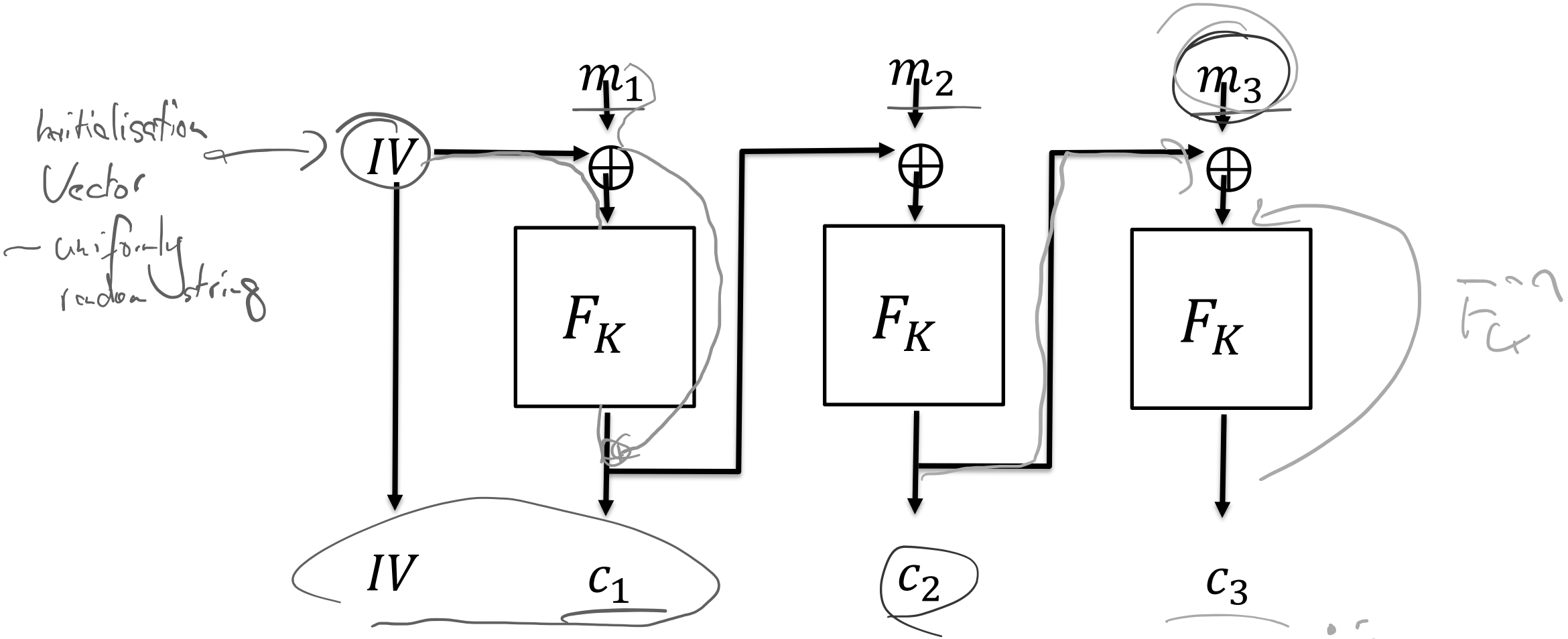
ECM



IND-secure



Cipher Block Chaining (CBC) Mode



- IND-CPA secure for uniformly random IV if F is PRP
- Encryption must be performed sequentially
- Decryption is local

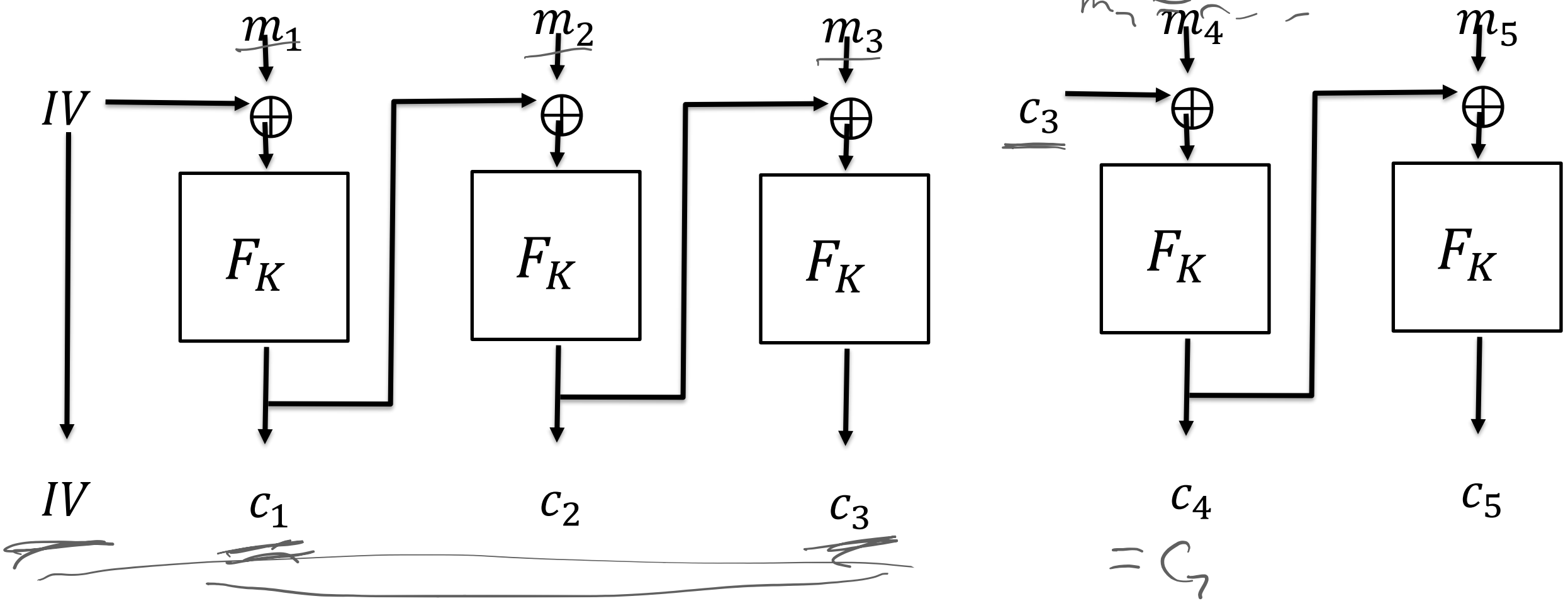
$(r, F_K(r) \oplus m)$

$(r, F_K(r \oplus m))$

Chained CBC Mode

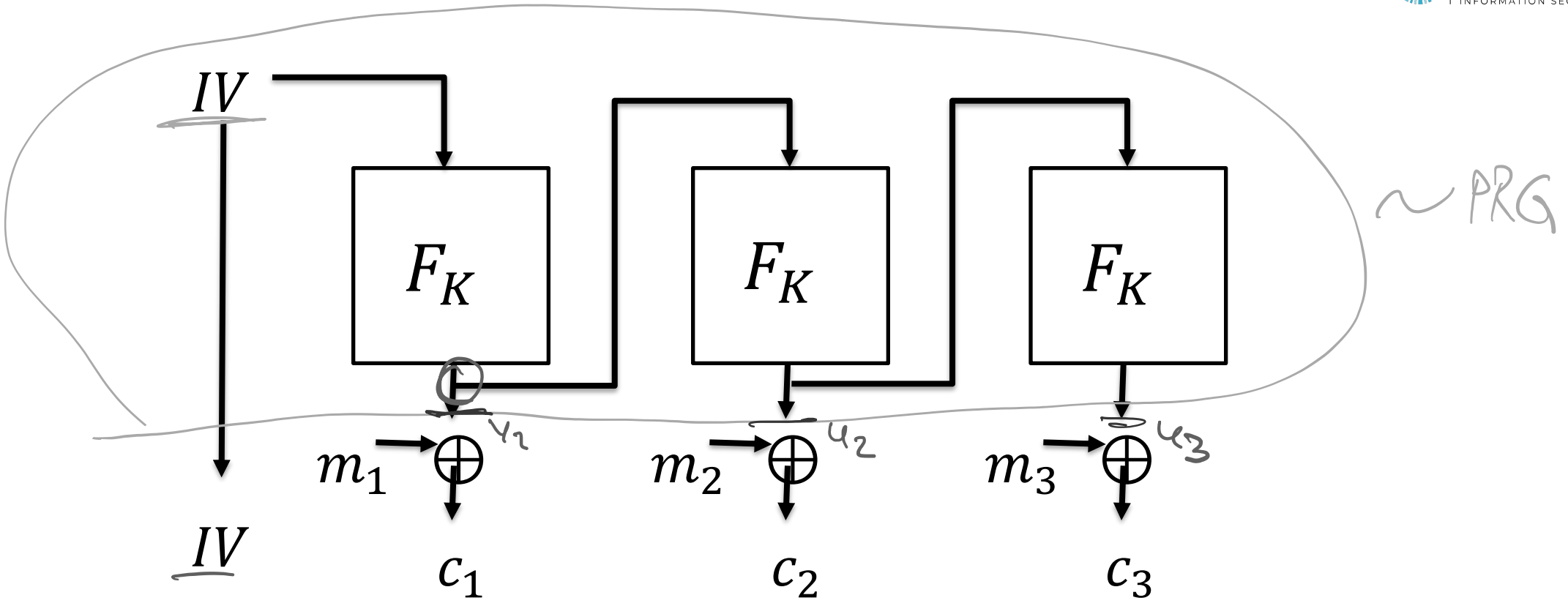
stateful

$m_0 = c_3 \oplus m_1 \oplus IV$



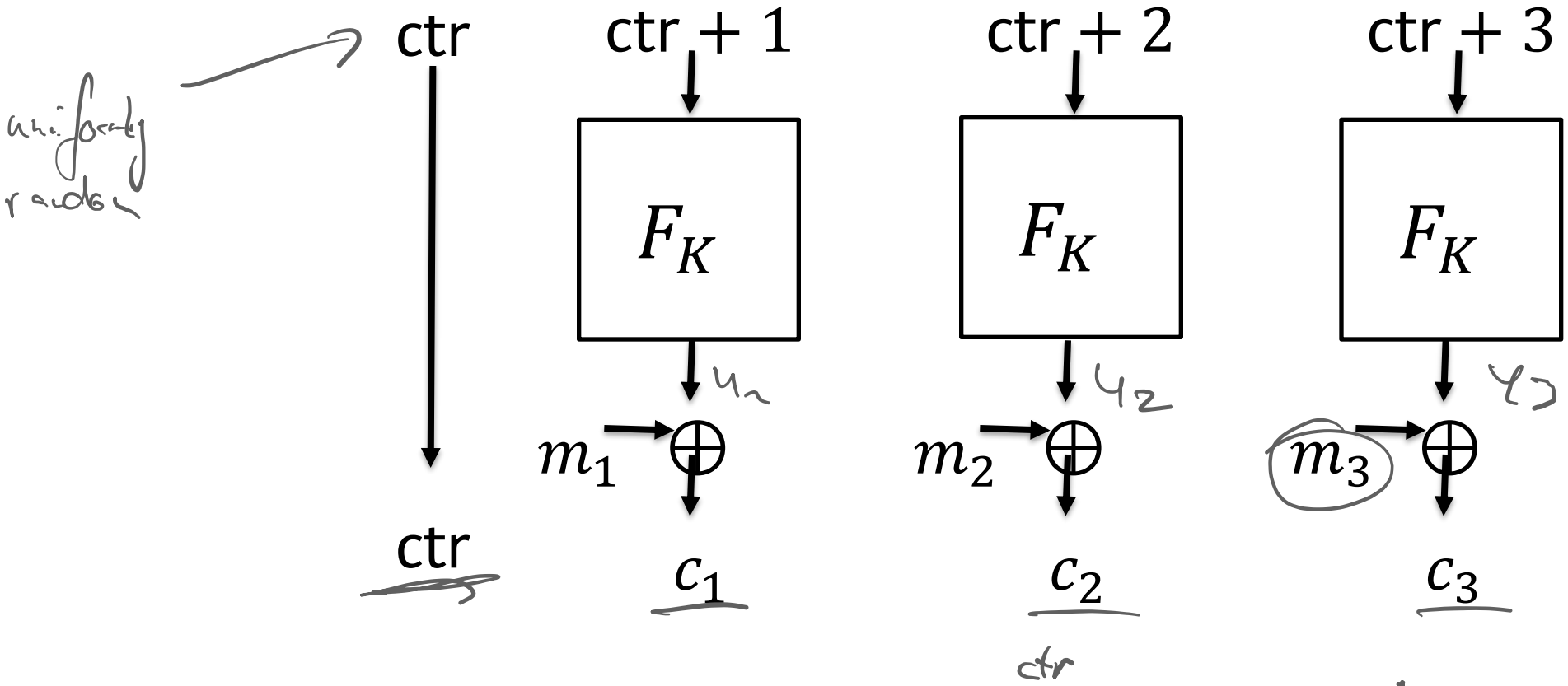
- **Not** IND-CPA secure!

Output Feedback (OFB) Mode



- IND-CPA secure for uniformly random IV if F is PRF
- Encryption and Decryption must be performed sequentially!
- Pad can be precomputed

Counter (CTR) Mode



- IND-CPA secure for uniformly random ctr if F is PRF
- Encryption and Decryption are local/can be parallelized/precomputed
- Not reason not to use this one!

Summary

- Blockcipher modes of operation provide a way to encrypt long messages
- ECM provides very little security → should only be used in very special scenarios
- CBC is ok!
- Chained CBC is not ok!
- OFM is ok!
- Use CTR whenever possible!